Assignment 8

October 11, 2026

Question 1

(a)

we are given:

- pressure: $P = 29\,\text{atm}$
- temperature: $T = 210°\text{C} = 483.15\,\text{K}$
- heat capacity ratio: $\gamma = 1.3$
- molar mass of ammonia (NH_3): $M = 17.03 \times 10^{-3}\,\text{kg mol}^{-1}$
- required resolution: $d = 1\,\text{mm}$

first we need the speed of sound in the ammonia atmosphere. treating ammonia as an ideal gas,

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

substituting,

$$v = \sqrt{\frac{(1.3)(8.314)(483.15)}{17.03 \times 10^{-3}}}$$

$$v \approx 554\,\text{m s}^{-1}$$

the smallest spot that can be resolved is given by Abbe's limit,

$$d = \frac{\lambda}{2\sin\theta}$$

where θ is the half angle of the diffracted waves collected by the alien's body. the alien is very close to the object ($\ll 1\,\text{m}$ while its body is about $1\,\text{m}$ across), so it collects waves from almost all angles up to

$$\theta \to 90°$$

$$\sin\theta \to 1$$

so,

$$d = \frac{\lambda}{2}$$
$$\lambda = 2d = 2(1 \times 10^{-3}) = 2 \times 10^{-3}\,\text{m}$$

using $v = f\lambda$,

$$f = \frac{v}{\lambda} = \frac{554}{2 \times 10^{-3}}$$
$$f \approx 2.77 \times 10^{5}\,\text{Hz}$$

$$\boxed{f \approx 277\,\text{kHz}}$$

so the alien needs to hear up to about 277 kHz.

(b)

now the object is 2 m away. the alien's body has diameter $D = 1\,\text{m}$, so it only collects waves up to a smaller angle.

$$\tan\theta = \frac{D}{2f_\text{d}}$$
$$\theta = \arctan\left(\frac{D}{2f_d}\right)$$

where $f_\text{d} = 2\,\text{m}$ is the distance to the object

$$\theta = \arctan\left(\frac{1}{2 \times 2}\right) = \arctan(0.25) \approx 14.04^\circ$$
$$\sin\theta \approx 0.2425$$

rearranging Abbe's limit,

$$\lambda = 2d\sin\theta$$
$$\lambda = 2(1 \times 10^{-3})(0.2425)$$
$$\lambda \approx 4.85 \times 10^{-4}\,\text{m}$$

then,

$$f = \frac{v}{\lambda} = \frac{554}{4.85 \times 10^{-4}}$$
$$f \approx 1.14 \times 10^{6}\,\text{Hz}$$

therefore,

$$\boxed{f \approx 1.1\,\text{MHz}}$$

Question 2

(a)

for a diffraction grating at normal incidence,

$$m\lambda = d\sin\theta$$

so for each order the diffraction angle is

$$\theta = \arcsin\left(\frac{m\lambda}{d}\right)$$

we want the angle between 400 nm and 700 nm to be as large as possible. using the first order ($m = 1$) for the best separation,

$$\Delta\theta = \arcsin\left(\frac{700\,\text{nm}}{d}\right) - \arcsin\left(\frac{400\,\text{nm}}{d}\right)$$

this increases as d gets smaller because the angles get larger. however, we need $\sin\theta \leq 1$ for the red light to be diffracted so

$$\frac{700\,\text{nm}}{d} \leq 1$$
$$d \geq 700\,\text{nm}$$

the best separation is at the smallest allowed period,

$$\boxed{d \approx 700\,\text{nm}}$$

(b)

using $d = 700\,\text{nm}$ and $m = 1$:

- for the 400 nm light,

$$\sin\theta_{400} = \frac{400}{700} = 0.5714$$
$$\theta_{400} \approx 34.85^\circ$$

- for the 700 nm light,

$$\sin\theta_{700} = \frac{700}{700} = 1$$
$$\theta_{700} = 90^\circ$$

the spectrum is therefore spread over

$$\Delta\theta = 90^\circ - 34.85^\circ$$
$$\boxed{\Delta\theta \approx 55^\circ}$$

from about 35° to 90°.

Question 3

(a)

the input splits into two waves that go round the loop in opposite directions,

$$E_A = tE_{in}e^{-i\phi_A} \qquad E_B = rE_{in}e^{-i\phi_B}$$

each output gets one transmitted and one reflected part,

$$E_1 = rE_A + tE_B \qquad E_2 = tE_A - rE_B$$

the optical path length is the same both ways, so $\phi_A = \phi_B = \phi$,

$$E_1 = 2rt\,E_{in}e^{-i\phi} \qquad E_2 = (t^2 - r^2)E_{in}e^{-i\phi}$$

for $r = t = \frac{1}{\sqrt{2}}$, $2rt = 1$ and $t^2 - r^2 = 0$,

$$\boxed{E_1 = E_{in}, \quad E_2 = 0}$$

(b)

the fibre length is $L = 2\pi NR$ and the coupler moves at ΩR. light travels at c/n.

let $t_0 = \dfrac{2\pi NnR}{c}$ (the travel time with no rotation) and $x = \dfrac{\Omega nR}{c} \ll 1$.

- clockwise (t_{CC}):

  the coupler moves away from the light, so in time t_{CC} the light must cover the fibre plus the extra distance the coupler moves,

$$\frac{c}{n}t_{CC} = 2\pi NR + \Omega R\,t_{CC}$$
$$t_{CC}\left(\frac{c}{n} - \Omega R\right) = 2\pi NR$$
$$t_{CC} = \frac{2\pi NR}{\frac{c}{n} - \Omega R}$$

  taking out $\frac{c}{n}$ from the bottom,

$$t_{CC} = \frac{2\pi NnR}{c}\,\frac{1}{1 - \frac{\Omega nR}{c}} = \frac{t_0}{1-x}$$

  since $x \ll 1$ we use $\frac{1}{1-x} \approx 1 + x$,

$$\boxed{t_{CC} \approx \frac{2\pi NnR}{c}\left(1 + \frac{\Omega nR}{c}\right)}$$

- anticlockwise (t_{AC}):
  the coupler moves towards the light, so the sign of Ω flips,

$$\frac{c}{n}t_{AC} = 2\pi NR - \Omega R\, t_{AC}$$
$$t_{AC} = \frac{2\pi NR}{\frac{c}{n} + \Omega R} = \frac{t_0}{1+x}$$

  using $\frac{1}{1+x} \approx 1 - x$,

$$\boxed{t_{AC} \approx \frac{2\pi NnR}{c}\left(1 - \frac{\Omega nR}{c}\right)}$$

  together these are the result we were asked to show,

$$t = \frac{2\pi NnR}{c}\left(1 \pm \frac{\Omega nR}{c}\right)$$

- difference:

$$\Delta t = t_{CC} - t_{AC} = t_0(1+x) - t_0(1-x) = 2xt_0$$

$$\Delta t = \frac{2\pi NnR}{c} \cdot \frac{2\Omega nR}{c} = \frac{4\pi N\Omega n^2 R^2}{c^2}$$

  the area inside the loop is $A = \pi R^2$, so

$$\boxed{\Delta t = \frac{4\Omega N n^2 A}{c^2}}$$

(c)

given:

- $\lambda = 1550$ nm
- $n = 1.5$
- $R = 1$ m
- $\Omega = 1\,\text{rev/hr} = \frac{2\pi}{3600} = 1.745 \times 10^{-3}\,\text{rad}\,\text{s}^{-1}$
- $r = t = \frac{1}{\sqrt{2}}$
- $A = \pi\,\text{m}^2$,

$$\Delta t = \frac{4(1.745\times10^{-3})(1000)(1.5)^2\pi}{(3.00\times10^8)^2} \approx 5.49\times10^{-16}\,\mathrm{s}$$

$$\omega = \frac{2\pi c}{\lambda} \approx 1.22\times10^{15}\,\mathrm{rad\,s^{-1}}$$

$$\Delta\phi = \omega\Delta t \approx 0.667\,\mathrm{rad}$$

for $r = t = \frac{1}{\sqrt{2}}$ the output is $E_2 = \frac{E_{in}}{2}\left(e^{-i\phi_A} - e^{-i\phi_B}\right)$, so

$$\frac{|E_2|^2}{E_{in}^2} = \sin^2\left(\frac{\Delta\phi}{2}\right) = \sin^2(0.3336) \approx 0.107$$

$$\boxed{\approx 11\%}$$